Solutions Manual to Accompany

Homework For Linear Control System Engineering

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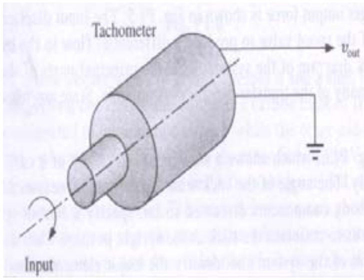
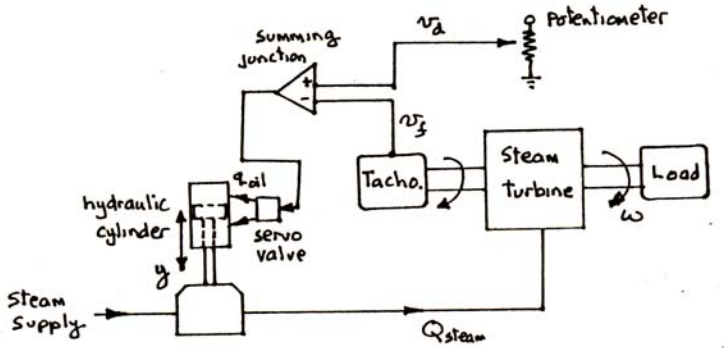
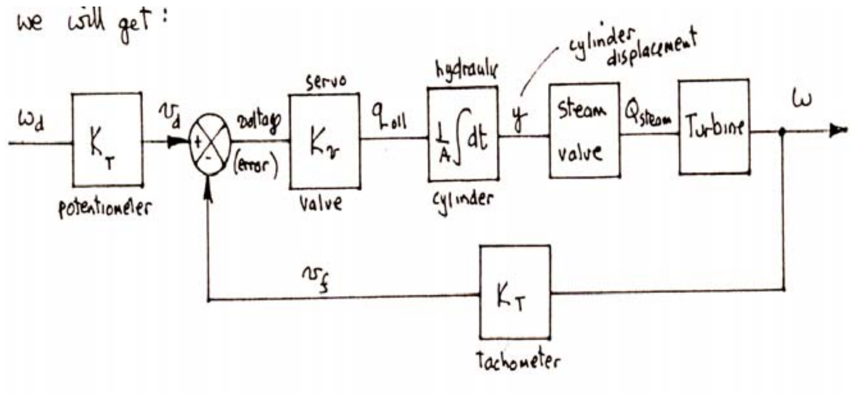
HIT 651

P1-2 Redesign the turbine speed control system discussed in Sample Problem 1.1, but replace the fly-ball governor with the tachometer shown in Fig.1.2. A tachometer consists basically of a small DC motor operated in reverse as a generator, the shaft being rotated continuously, producing a DC voltage proportional to shaft speed.

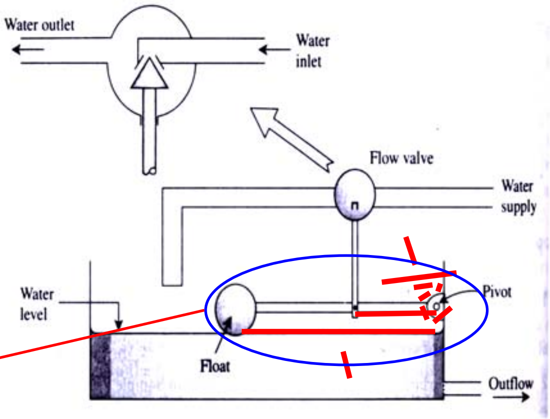


Solution

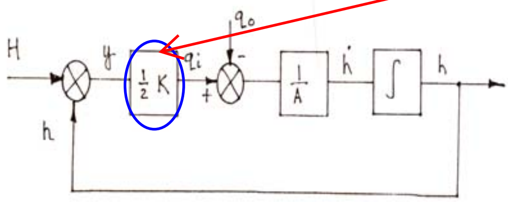




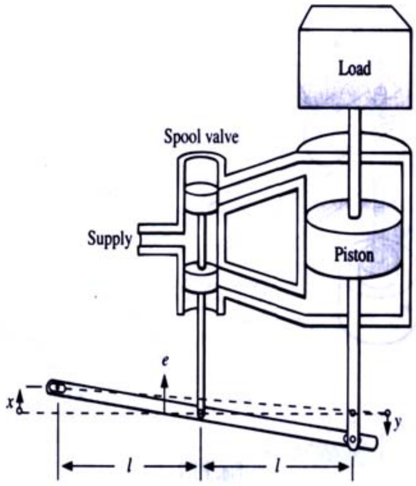
P1-4. Shown in Fig.P1.4 is a water-level control system comprising a tank, inlet pipe, slide valve, and float. Details of the operation of the flow valve are also shown in the figure. Draw a block diagram of the feedback control system and identify the main elements, writing down the mathematical transfer functions where appropriate.



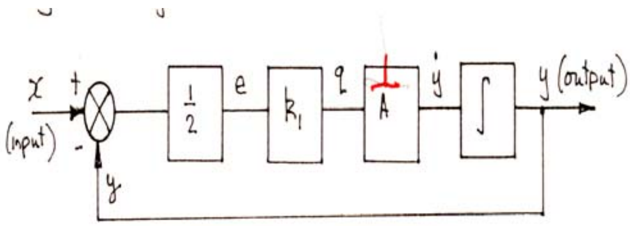
Solution



P1-5 A method of producing a displacement proportional to an input displacement but with a much larger output force is shown in Fig.P1.5. The input displacement x causes the movement of the spool valve to produce a differential flow to the hydraulic actuator. Draw a block diagram of the system, label the principal parts of the control loop, and identify as many of the transfer functions as possible. State any assumptions made in the analysis.



Solution



P2-1 A system unknown transfer function is shown in Fig.P2.1. If a unit impulse applied at the input produces at the output a signal described by the time function $c(t) = 2e^{-3t}$, determine the unknown transfer function.

Solution

$$c(t) = 2e^{-3t} \quad => \quad \frac{C}{R} = \frac{2}{s+3}$$

P2-2. Find the solution of the differential equation

$$\frac{d^2x}{dt^2} + \frac{dx}{dt} + 8x = \frac{dz}{dt} + 3z$$

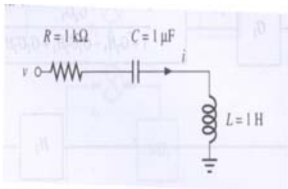
When $z(t) = e^{-2t}$ and all other initial conditions are zero.

Solution $Z(s) = \dfrac{1}{s+2} \cdot \dfrac{s+3}{s^2+s+8} = \left(\dfrac{1}{s+2} - \dfrac{s+C_2}{s^2+s+8}\right) \cdot C_1 = \left(\dfrac{1}{s+2} - \dfrac{s-11}{(s+\frac{1}{2})^2 + \frac{31}{4}}\right) \cdot \dfrac{1}{10}$

$$y(t) = 0.1e^{-2t} - 0.1e^{-\frac{1}{2}t}\left(\cos\frac{\sqrt{31}}{2}t - \frac{23}{\sqrt{31}}\sin\frac{\sqrt{31}}{2}t\right)$$

P2-3 For the system shown in Fig.P2.3., determine the relationship between voltage and current, express this relationship in the form of a transfer function and determine the current as a function of time when the voltage is a step change from zero to 10V.

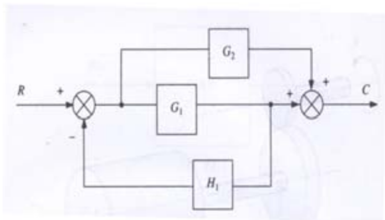
Solution $\dfrac{1}{R + sL + 1/Cs} = \dfrac{Cs}{LCs^2 + RCs + 1}$,



$$Y(s) = \frac{10}{s}\frac{Cs}{LCs^2 + RCs + 1} = \frac{10 \cdot 10^{-6}}{10^{-6}s^2 + 10^{-3}s + 1} = \frac{10}{s^2 + 10^3 s + 10^6}$$

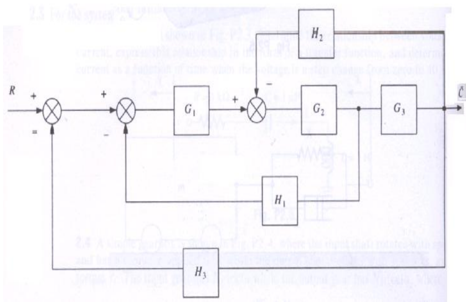
$$y(t) = 10^{-2}\frac{2}{\sqrt{3}}e^{-500t}\sin(\frac{\sqrt{3}}{2}1000t) = 0.01154 \cdot e^{-500t}\sin(866t)$$

P2-8 For the system shown in Fig2.8 determine the closed –loop transfer function C/R.



Solution $\dfrac{C}{R} = \dfrac{G_1}{1+G_1H} \cdot \left(1 + \dfrac{G_2}{G_1}\right) = \dfrac{G_1 + G_2}{1+G_1H}$

P2-9 For the single input system shown in Fig2.9, find the transfer function of output to input C/R.



Solution $\dfrac{G_1G_2}{1+G_2G_3H_2} \quad => \quad \dfrac{G_1G_2G_3}{1+G_2G_3H_2 + G_1G_2H_1}$

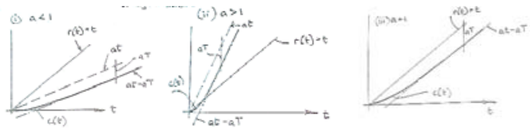
$$\frac{C}{R} = \frac{G_1G_2G_3}{1+G_2G_3H_2 + G_1G_2H_1 + G_1G_2G_3H_3}$$

P3.2 Determine the output of the open-loop system G(s)=a/(1+sT) to the input r(t)=t. Sketch both input and output as function of time, and determine the steady-state error between the input and output. Compare the result with that given by Fig.3.7.

Solution: $c(s) = R(s)\cdot\dfrac{a}{(1+sT)} = \dfrac{a}{s^2\cdot(1+sT)}$ $\quad c(t) = a\cdot(t - T + Te^{-t/T})$

$e = r - c = t - a(t + T - Te^{-t/T})$

As $t\to\infty$, $e_{ss} = t - at - aT = \begin{cases} T & \text{for} \quad a = 1 \\ \infty & \text{for} \quad a \neq 1 \end{cases}$



P3.5 An open-loop first-order system is characterized b the transfer function $G(S) = \dfrac{1}{1+\tau s}$, where the time constant is $\tau = 5s$. Calculate the steady-state error when the system input is r(t)=1+6t. Confirm the result by using the final-value theorem.

Solution: By the superposition theorem, the system output c(t) could be considered as a sum of a step response and a ramp response. That is

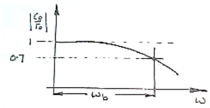
$$e(t) = r - c = (6t+1) - 6\cdot(t - 5 + e^{-t/5}) - (1 - e^{-t/5}) = 30 - 5e^{-t/5}, \quad \text{yields} \quad e(\infty) = 30$$

By the final value theorem for Laplace's transform,

$$E(s) = R(s) - C(s) = R(s)\cdot[1 - G(s)] = (\frac{6}{s^2} + \frac{1}{s})\cdot\left(1 - \frac{1}{5s+1}\right) = \frac{(6+s)\cdot 5}{s\cdot(5s+1)}$$

$$\lim_{t\to\infty} e(t) = \lim_{s\to 0} s\cdot E(S) = \lim_{s\to\infty} s\cdot\frac{(6+s)\cdot 5}{s\cdot(5s+1)} = 30$$

P3.7 One definition of the bandwidth of a system is the frequency range over which the amplitude of the output signal is greater than 70% of the input signal amplitude when a system is subjected to a harmonic input. Find a relationship between the bandwidth and time constant of a first-order system. What is the phase angle at the bandwidth frequency?



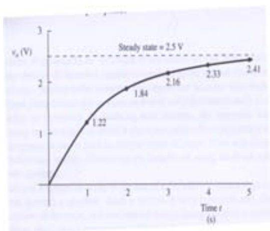
Solution: $\left|\dfrac{1}{1+\tau s}\right| = \dfrac{1}{\sqrt{1+\omega_b^2\tau^2}} = 0.707 \approx \dfrac{\sqrt{2}}{2}$ or $1 + \omega_b^2\tau^2 = 2$

i.e. $\omega_b = 1/\tau$ $\quad \angle G(j\omega) = -\tan^{-1}(\omega\tau) = -\tan^{-1}(1) = -45°$, phase lag is $45°$ at the bandwidth frequency.

P3.8 Figure P3.8 shows the experimentally obtained voltage output of an unknown system subjected to a step input of +10V. Determine the transfer function of the system and locate its pole on the complex plane.

Solution: System appears to be first order. So suppose the transfer function is as follow:

$G(s) = \dfrac{C}{R} = \dfrac{K}{1+\tau s}$, then $c(t) = 10K\cdot(1 - e^{-t/\tau})$



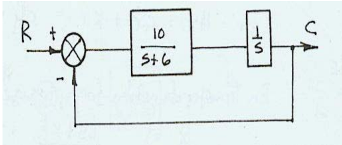
From final value theory

$\lim_{t\to\infty} c(t) = \lim_{s\to 0} s\cdot C(s) = \lim_{s\to 0} s\cdot\dfrac{10}{s}\dfrac{K}{\tau s+1} = 10\cdot K = 2.5V$, hence K=0.25

Now consider time constant, $\tau = t\cdot\dfrac{1}{-\ln\left(1 - \dfrac{1}{10K}c(t)\right)}$ =1.4983

Matlab: K=0.25, c=[1.22, 1.84, 2.16, 2.33, 2.41], t=1:5, tao=t./(-log(ones(1,5)-C/10/K))*ones(5,1)/5

P4. 1 Figure P4.1 shows a closed-loop feedback system with a second-order plant. Determine the damped natural frequency and damping ratio of the closed-loop response.

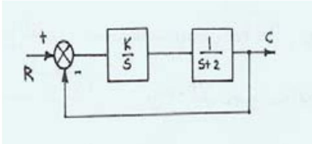


$$\frac{C(s)}{R(s)} = \frac{G}{1+GH} = \frac{10}{s^2+6s+10} = \frac{\omega_n^2}{s^2+2\zeta\omega_n s+\omega_n^2}$$

Hence $\omega_n = \sqrt{10} = 3.16$ rad/s, $\zeta\omega_n = 6/2 = 3$ => $\zeta = 3/\sqrt{10} = 0.3\sqrt{10}$

$$\omega_d = \omega_n\sqrt{1-\zeta^2} = \sqrt{10}\cdot\sqrt{1-0.9} = 1 \text{ rad/s}$$

P4. 4 Calculate the required value of gain K shown in follow such that the closed-loop response of the system to a step input is limited to no more than 10% overshoot. Plot the closed-loop poles on the complex plane for the same value of gain.



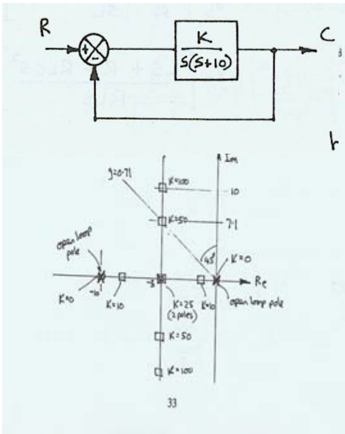
$$\frac{C(s)}{R(s)} = \frac{G}{1+GH} = \frac{K}{s^2+2s+K} = \frac{\omega_n^2}{s^2+2\zeta\omega_n s+\omega_n^2} \qquad \zeta\omega_n = 2/2 = 1, \quad \omega_n^2 = K$$

By textbook fig. 4. 17, 10% overshoot is responding to $\zeta = 0.6$,

$$\omega_n = \frac{1}{\zeta} = \frac{5}{3} = 1.67, \quad K = \frac{25}{9} = 2.78, \quad \omega_d = \omega_n\sqrt{1-\varsigma^2} = \frac{5}{3}\sqrt{1-0.6^2} = \frac{4}{3} = 1.33$$

The closed-loop poles will be $s^2+2s+\frac{25}{9} = 0$ => $s = -1 \pm j\frac{4}{3}$ or $s = -\varsigma\omega_n \pm j\omega_d = -1 \pm j1.33$

P4. 5 A unity-feedback control system has the forward-path transfer function G(s)=K/s(s+10). Find the closed –loop transfer function, and develop expression for the damping ratio and damped natural frequency in terms of K. Plot the close-loop poles on the complex plane for K=0,10,25,50,100. For each value of K calculate the corresponding damping ratio and damped natural frequency. What conclusions can you draw from the plot?



$$\frac{C(s)}{R(s)} = \frac{K}{s^2+10s+K} = \frac{\omega_n^2}{s^2+2\zeta\omega_n s+\omega_n^2}$$

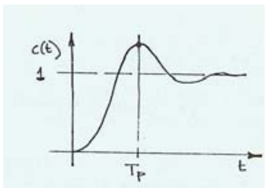
Hence $\omega_n = \sqrt{K}$ and $2\zeta\omega_n = 10$ or $\zeta = \frac{5}{\sqrt{K}}$

Closed loop poles located at

$$s = -\zeta\omega_n \pm j\sqrt{1-\varsigma^2}\,\omega_n = -5 \pm j\sqrt{K-25}$$

P4. 7 Prove that for an under damped second-order system subject to a step input, the percentage overshoot above the steady-state output (as shown in textbook Fig.P4.7) is a function only of the damping ratio ζ.

By step response $c(t) = 1 - \frac{1}{\sqrt{1-\zeta^2}} e^{-\zeta\omega_n t}\sin(\omega_d t+\phi)$



At the peak time, $-\zeta\omega_n e^{-\zeta\omega_n t}\sin(\omega_d t+\phi) + \omega_d e^{-\zeta\omega_n t}\cos(\omega_d t+\phi) = 0$

$$tg(\omega_d t+\phi) = \omega_d/\varsigma\omega_n = \phi \Rightarrow \omega_d T_p = 0, \pi, 2\pi... \Rightarrow T_p = \frac{\pi}{\omega_d}$$

$$PO = \frac{1}{\sqrt{1-\zeta^2}} e^{-\zeta\omega_n\frac{\pi}{\omega_d}} \sin(\omega_d\frac{\pi}{\omega_d}+\phi) = \frac{1}{\sqrt{1-\zeta^2}} e^{\frac{-\zeta\pi}{\sqrt{1-\zeta^2}}}\sin\phi = e^{\frac{-\zeta\pi}{\sqrt{1-\zeta^2}}}$$

P5. 2 A second-order system is given by the transfer function

$$\frac{C(s)}{R(s)} = \frac{10}{s^2+2s+9} = \frac{10}{9}\frac{9}{s^2+2s+9}$$

Determine:

a. damping ratio

b. The damped and undamped natural frequency.

c. Maximum peak modulus Mp

d. The frequency at which Mp occurs.

e. The bandwidth, defined as the frequency range over which the modulus does not fall more then 3db below the low-frequency (DC) value.

f. The steady-state output for a unit step input.

g. The location of the closed-loop poles on the complex plane.

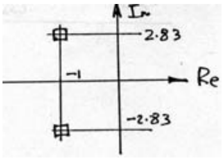
h. The rise time.

i. The 2% settling time

Solution:

$\omega_n = 3\text{rad/s}$ and $\zeta\omega_n = 1$, so $\zeta = 1/3$, $\omega_d = \sqrt{\omega_n^2 - (\zeta\omega_n)^2} = \sqrt{9-1} = 2\sqrt{2} = 2.83\text{rad/s}$

$M_p = \dfrac{10/9}{2\varsigma\cdot\sqrt{1-\zeta^2}} = \dfrac{10}{2\sqrt{8}} = \dfrac{5}{4}\sqrt{2} = 1.77$, $\omega_p = \omega_n\sqrt{1-2\zeta^2} = \sqrt{7}$ =2.65



From $M = 0.707 = \dfrac{1}{\sqrt{(1-r^2)^2 + 4\zeta^2 r^2}}$, where $r=\omega/\omega_n$

Squaring, and setting $x = r^2$, gives $x^2 - 2x + 1 + (4/9)x = 2$ => $x^2 - (14/9)x - 1 = 0$

$$x=\frac{7}{9} \pm \frac{\sqrt{49+81}}{9} = \frac{7\pm\sqrt{130}}{9}, \quad r = \frac{1}{3}\sqrt{7+7.07} = \frac{4.29}{3}, \qquad \omega_b = r\cdot\omega_n = 4.29$$

f. R(s)=1/s, $e_{ss} = \lim\limits_{t\to\infty} c(t) = \lim\limits_{s\to 0} s\cdot\dfrac{1}{s}\cdot\dfrac{10}{s^2+2s+9} = \dfrac{10}{9}$

g. closed-loop poles for s^2+2s+9, $s = -1 \pm j2\sqrt{2}$

h. Rise time $T_r = \dfrac{1}{\omega_d}\left(\pi - \tan^{-1}\dfrac{\sqrt{1-\zeta^2}}{\zeta}\right) = \dfrac{1}{2\sqrt{2}}\cdot 1.91\text{rad} = 0.68\text{s}$

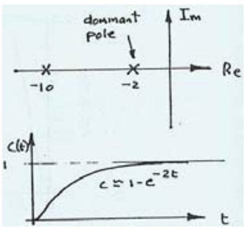
i. 2% settling time $T_s = \dfrac{4}{\zeta\omega_n} = 4\,\text{sec}$

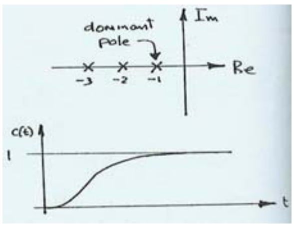
P5.3 Consider each of the following closed-loop transfer functions. By considering the location of the poles on the complex plane, sketch the unit step response, explaining the results obtained:

Solution:

a. $\dfrac{C(s)}{R(s)} = \dfrac{20}{(s+2)(s+10)}$

$e_{ss}= \lim\limits_{t\to\infty} c(t) = \lim\limits_{s\to 0} s\cdot\dfrac{1}{s}\cdot\dfrac{20}{(s+2)(s+10)} = 1$

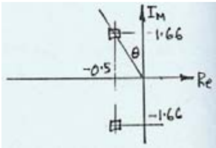


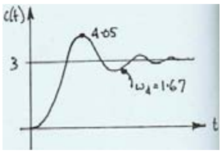


b. $\dfrac{C(s)}{R(s)} = \dfrac{6}{(s+1)(s+2)(s+3)}$

Dominant time constant=1s, $e_{ss} = \lim\limits_{s\to 0} s\cdot\dfrac{1}{s}\cdot\dfrac{6}{(s+1)(s+2)(s+3)} = 1$

c. $\frac{C(s)}{R(s)}=\frac{9}{s^2+s+3}$





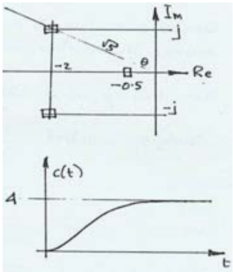
Poles: $s=-0.5\pm j0.5\sqrt{11}$

$$\omega_n=\sqrt{3},\zeta=0.5/1.732=0.289$$

from Fig4.7, $PO=35\%$, $\Delta c=4.05$

$$e_{ss}=\lim_{s\to0}s\cdot\frac{1}{s}\cdot\frac{9}{s^2+s+3}=3$$

d. $\frac{C(s)}{R(s)}=\frac{10}{(s^2+4s+5)(s+0.5)}$

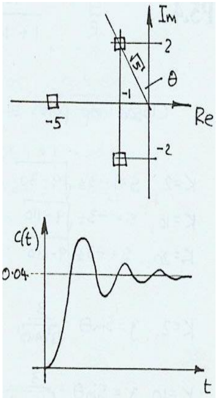


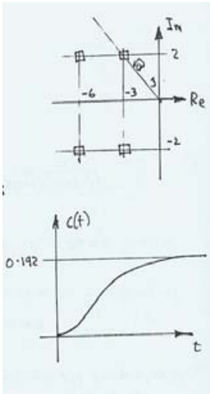
Solving $s^2+4s+5=0$ gives $s=-2\pm j$

$$\zeta=\sin\theta=\frac{2}{\sqrt{5}}=0.4\sqrt{5}=0.89$$

No overshoot, however, real axis pole will dominate with a time constant at 2s. e_{ss}=10/2.5=4

e. $\frac{C(s)}{R(s)}=\frac{1}{(s+5)(s^2+2s+5)}$





Solving $s^2+2s+5=0$

gives $s=-1\pm j2$ $\quad\zeta=\sin\theta=\frac{1}{\sqrt{5}}=0.45$

The complex pair will dominate the real axis pole,
e_{ss}=1/25=0.04

f. $\frac{C(s)}{R(s)}=\frac{100}{(s^2+6s+13)(s^2+12s+40)}$

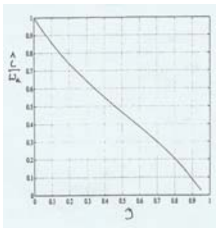
Solving $s^2+6s+13$, gives $s=-3\pm 2j$

Solving $s^2+12s+40$, gives $s=-6\pm 2j$

Complex pair closer to imaginary axis will dominate those further away.

$$\zeta_{dom}=\frac{3}{\sqrt{13}}=0.83$$ High damping, no or little overshoot, Steady state output=100/(13*40)=0.192

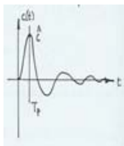
P5.6 Determine the magnitude of the first overshoot of an under damped, second-order system subjected to a unit impulse input. Plot a graph of this magnitude as a function of damping ratio ζ.



Solution: $\frac{C(s)}{R(s)}=\frac{\omega_n^2}{s^2+2\zeta\omega_n+\omega_n^2}$ If R(s)=1 then $C(s)=\frac{\omega_n^2}{(s+\zeta\omega_n)^2+\omega_d^2}$

$$c(t)=\frac{\omega_n^2}{\omega_d}e^{-\zeta\omega_n t}\sin\omega_d t \quad \text{where} \quad \omega_d=\omega_n\sqrt{1-\zeta^2}$$

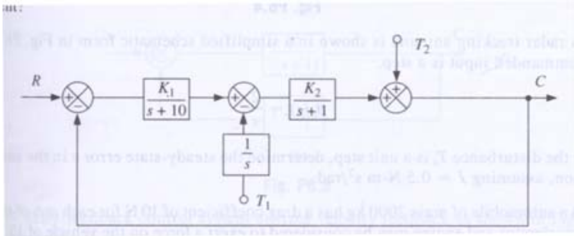
let $\frac{dc(t)}{dt}=0$, $-\zeta\omega_n e^{-\zeta\omega_n t}\sin\omega_d t+\omega_d e^{-\zeta\omega_n t}\cos\omega_d t=0$ hence $T_p=\frac{\pi}{\phi}$



at this time ,output $M=\frac{\omega_n^2}{\omega_d}e^{-\varsigma\omega_n\frac{\phi}{\omega_d}}\sin\omega_d\frac{\phi}{\omega_d}=\omega_n\exp\left(-\frac{\varsigma}{\sqrt{1-\zeta^2}}tg^{-1}\frac{\sqrt{1-\zeta^2}}{\varsigma}\right)$

This is a function of ω_n and ζ, so plot M/ω_n against ζ.

P6.1 Figure P6.1 shows a unity-feedback control system with one input R and two disturbance T1 and T2. Find the transfer function relating the output to each of the three inputs and determine the characteristic equation. What is the output if all inputs are present?



Solution:

$$\frac{C(s)}{R(s)}=\frac{K_1K_2}{K_1K_2+(s+1)(s+10)}=\frac{K_1K_2}{s^2+11s+(10+K_1K_2)}$$

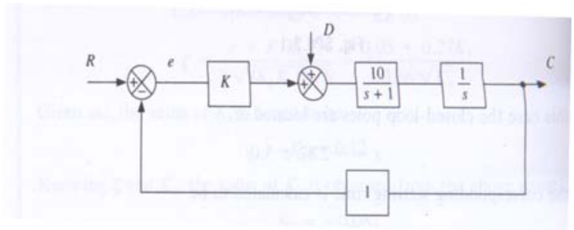
$$\frac{C(s)}{T_1(s)}=-\frac{1}{s}\cdot\frac{K_2\cdot(s+10)}{K_1K_2+(s+1)(s+10)}=\frac{-K_2(s+10)}{s\cdot(s^2+11s+10+K_1K_2)}$$

$$\frac{C(s)}{T_2(s)}=\frac{(s+1)(s+10)}{K_1K_2+(s+1)(s+10)}=\frac{(s+1)(s+10)}{s^2+11s+(10+K_1K_2)}$$

The characteristic equation is $s^2+11s+(10+K_1K_2)=0$ or $s[s^2+11s+(10+K_1K_2)]=0$.

$$C(s)=\frac{K_1K_2}{s^2+11s+10+K_1K_2}R(s)-\frac{K_2\cdot(s+10)}{s\cdot(s^2+11s+10+K_1K_2)}T_1(s)+\frac{(s+1)(s+10)}{s^2+11s+10+K_1K_2}T_2(s)$$

P6.3 A unity-feedback control system with a single disturbance input is shown in Fig.P6.3. Calculate the transfer function relating the error e to the two inputs R and D. Calculate the steady-state error e_{ss} when r(t) is a unit ramp. d(t) is a unit step, and K=10.



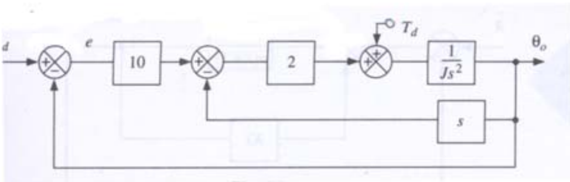
Solution:

$$\frac{E(s)}{R(s)}=\frac{s(s+1)}{s(s+1)+10K}=\frac{s(s+1)}{s^2+s+10K},$$

$$\frac{E(s)}{D(s)}=-\frac{10}{s(s+1)+10K}=-\frac{10}{s^2+s+10K}$$

$$e_{ss}=\lim_{t\to\infty}e(t)=\lim_{s\to0}s\cdot\frac{1}{s^2}\cdot\frac{s(s+1)}{s^2+s+10K}-\lim_{s\to0}s\cdot\frac{1}{s}\cdot\frac{10}{s^2+s+10K}=\frac{1-10}{10K}=\frac{-9}{100}=-0.09$$

P6.5 A radar-tracking antenna is shown in a simplified schematic form in Fig.P6.5. If the commanded input is a step, $\theta_d(s)=2/s$. While the disturbance T_d is a unit step, determine the steady-state error e in the antenna position, assuming J=0.5 N-m s^2/rad.



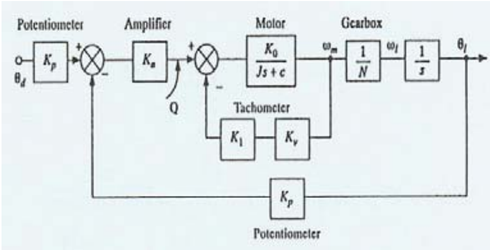
Solution:

$$\frac{E(s)}{\theta_d(s)}=\frac{20\cdot(Js^2+2s)}{(Js^2+2s)+20}=\frac{20\cdot s\cdot(Js+2)}{Js^2+2s+20},$$

$$\frac{E(s)}{\tau_d(s)}=-\frac{1}{Js^2+2\cdot(s+10)}=-\frac{1}{Js^2+2s+20}$$

$$e_{ss}=\lim_{s\to0}s\cdot\frac{2}{s}\cdot\frac{20s\cdot(0.5s+2)}{0.5s^2+2s+20}-\lim_{s\to0}s\cdot\frac{1}{s}\cdot\frac{1}{0.5s^2+2s+20}=-0.05$$

P6.10 Consider the system discussed in SP6.3. Represent graphically, in as compact a form as possible, the relationship between ς , ω_n and the system parameters K_a and K_l. What conclusions may be drawn from the graph regarding the possible range of ς . If the sign of the rate feedback signal is changed from a minus to a plus, what effect does this have on the graph?



Solution:

$$\frac{\theta_l}{\theta_d}=\left[\frac{K_pK_aK_0/N}{s\cdot(Js+c+K_1K_vK_0)+K_pK_aK_0/N}\right]=\frac{K}{Js^2+Bs+K}=\frac{K/J}{s^2+(B/J)s+(K/J)}$$

$$K=K_pK_aK_0/N \quad , \quad B=c+K_1K_vK_0$$

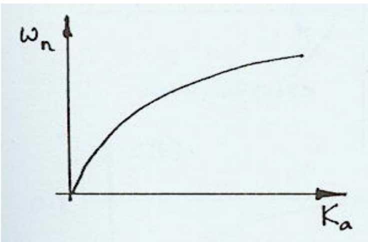
$$\omega_n=\sqrt{K/J}=\sqrt{K_pK_aK_0/NJ} \quad =\sqrt{6\cdot 13.5/(30\cdot 0.0125)}\sqrt{K_a}=14.7\sqrt{K_a}$$

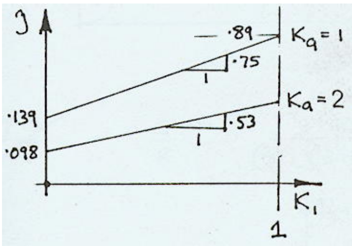
$$\varsigma=\frac{B/J}{2\omega_n}=\frac{c+K_1K_vK_0}{2\cdot J\cdot\omega_n} \quad =\frac{0.05+0.02\cdot 13.5\cdot K_1}{2\cdot 0.0125\cdot 14.7\sqrt{K_a}}=\frac{0.05+0.27K_1}{0.3675\cdot\sqrt{K_a}}$$

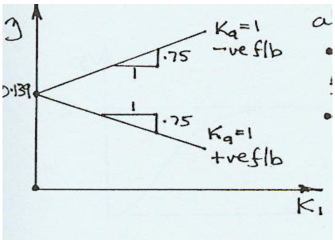
Clearly, only K_a affects ω_n. As for ζ , it is likely that K_1 will affect it the most, so we will plot graph for $\zeta-K_1$ with fixed values of K_a. Suppose Ka=1, then $\zeta=0.139+0.75K_1$. So that if $0<K_1<1$ then $0.139<\zeta<0.89$. Now if K_a=2, then $\zeta=0.098+0.53K_1$, and so on.

A conclusion is B to be only dependent K_l and independent of K_a , but ς will be varied with either K_a or K_1 .

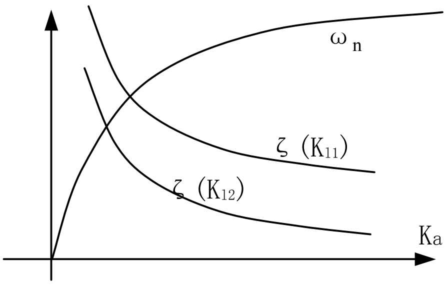
If the rate feedback signs changes, it is the same as making $K_1<0$. That is say $\zeta=0.139-0.75K_1$ (as K_a=1) . It means the value of ζ could be decreased by K_1.







In order to represent the relationship between ς , ω_n and the system parameters K_a and K_l graphically in as compact a form as possible, several curves could be drawn in one graph.



For problems 7.1-7.4 reduce the transfer function to an approximate first of second-order system. Plot the closed-loop poles on the complex plane, and sketch the unit step response.

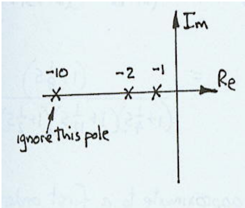
P7.2 $\dfrac{C}{R} = \dfrac{100}{(s+1)(s^2+12s+20)}$

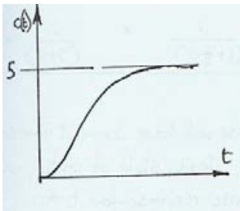
Solution: $\dfrac{C}{R} = \dfrac{100}{(s+1)(s+2)(s+10)} = \dfrac{5}{(1+s)(1+\frac{1}{2}s)(1+\frac{1}{10}s)} \cong \dfrac{5}{(1+s)(1+\frac{1}{2}s)}$ $\quad C_{ss} = 5$

That is a second-order, over damped system.

Further reduction might be $\dfrac{C}{R} \cong \dfrac{5}{(1+s)}$

first order , with dominant time constant $\tau = 1$.





P7.6 For the system shown in

$$\frac{C}{R} = \frac{10}{(s+1)(s^2+2s+6)}$$

Determine the maximum value of the output when the input takes the form R(s)=4/s

Solution: Since $\varsigma\omega_n = 1$ and $\tau = 1$, then $\beta = 1$, hence the percentage overshoot could be negligible. So the maximum output should be the steady state output.

If R=4/s, then $c_{max} = c(\infty) = 4 \cdot \dfrac{10}{6} = 6.67$

P7.8 Calculate the maximum value of output when a unit step input is applied to the system described by the closed-loop transfer function

$$\frac{C}{R} = \frac{4(s+3)}{2s^2+3s+10}$$

Solution: $\varsigma\omega_n = \dfrac{3/2}{2} = 0.75$, $\quad 1/\tau = 3 \quad \Rightarrow \quad \gamma = \dfrac{1/\tau}{\varsigma\omega_n} = \dfrac{3}{0.75} = 4$

By $\omega_n = \sqrt{10/2} = \sqrt{5}$, yields $\varsigma = 0.75/\sqrt{5} = 0.34$

From textbook Fig7.3 , when $\varsigma = 0.34$ and $\gamma = 4$, the percentage overshoot should be 45%.

The maximum value of output with unit step input is $\dfrac{12}{10} \cdot (1+0.45) =$ 1.74

P8.1 For unity-feedback systems described by the following open-loop transfer functions, determine the system type, any finite, nonzero error coefficients, and the corresponding steady-state error. $H(s) = 1$

(a) $G(s) = \dfrac{10}{s^2}$ $\therefore$ Type 2 system, $K_a = \lim\limits_{s\to 0} s^2 \cdot G(s) = 10$,

steady- state error to a unit acceleration input is $e_{ss}(\infty) = \dfrac{1}{K_a} = 0.1$

(b) $G(s) = \dfrac{s+1}{s(s^3 + 2s^2 + 7s + 5)}$ type 1 system, $K_v = \lim\limits_{s\to 0} s \cdot G(s) = 0.2$

steady state error to a unit ramp input is $e_{SS}(\infty) = \dfrac{1}{K_V} = 5$

(c) $G(s) = \dfrac{7}{s^5 + 2s^4 + 3s^3 + 3s^2 + 2s + 1}$ type 0 system, $K_p = \lim\limits_{s\to 0} G(s) = 7$

steady state error to a unit step input is $e_{ss}(\infty) = \dfrac{1}{1+Kp} = 0.125$

(d) $G(s) = \dfrac{5}{(s+1)^3}$ type 0 system, $K_p = \lim\limits_{s\to 0} G(s) = 5$

steady state error to a unit step input is $e_{ss}(\infty) = \dfrac{1}{1+Kp} = 0.167$

(e) $G(s) = \dfrac{(s+2)(s+3)(s^2+4)}{s}$ type 1 system, $K_v = \lim\limits_{s\to 0} s \cdot G(s) = 24$

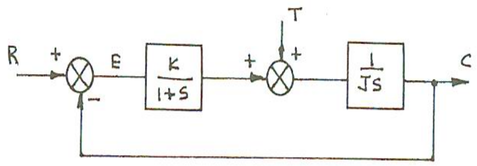
steady state error to a unit ramp input is $e_{ss}(\infty) = \dfrac{1}{Kv} = 0.0417$

P8.8

Figure P8.8 shows a speed control system subject to a disturbance torque T. If the system transfer functions are given by Gc(s)=K/(s+1), H(s)=1.

Determine the steady-state error due to a unit step change in input R (assuming T constant) and a unit step change in the disturbance torque T(assuming R constant), assuming K=1.

Determine K such that the resulting change in speed is less than 10% of the disturbance value.



Since System is type I, so the steady-state error due to a unit step change in input R is zero.

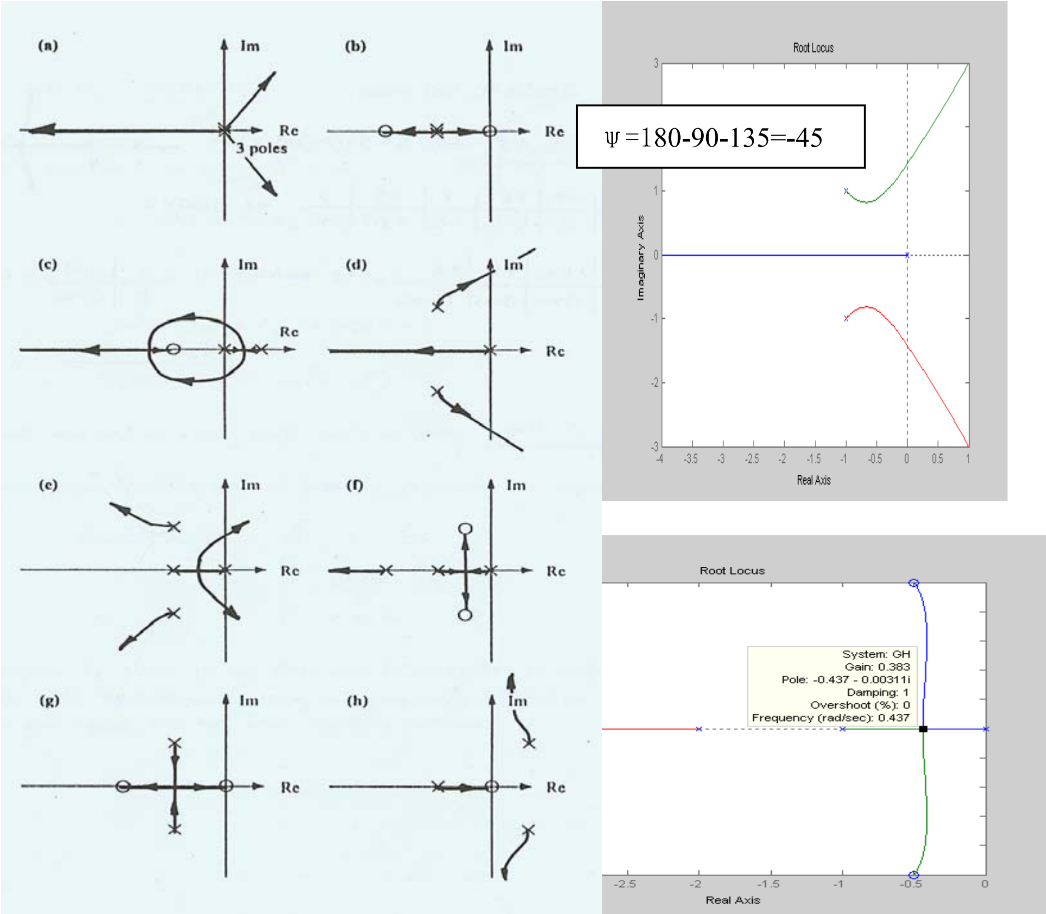
As R=0, $\dfrac{E}{T} = -\dfrac{s+1}{Js\cdot(s+1)+K}$

so $e_{SS} = \lim\limits_{t\to 0} e(t) = \lim\limits_{s\to 0} s\cdot\dfrac{1}{s}\cdot\dfrac{-(s+1)}{Js\cdot(s+1)+K} = -\dfrac{1}{K}$

when K=1, e_{ss}=1

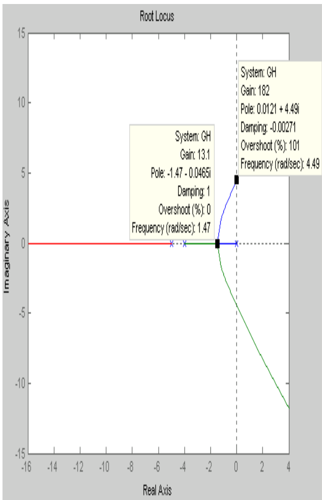
Obviously, as e_{ss}=10% of disturbance change value, the gain K should be increased to K=10

P9.6 For each of the open-loop poles and zeros shown in Fig.P9.6, sketch the corresponding root locus.



P9.7 The open-loop pole-zero map of a satellite attitude control system is shown in Fig.P9.7. Draw the root locus and determine, as accurately as possible, the point at which the locus leaves the real axis.

Solution: $GH(s) = \dfrac{1}{s \cdot (s+4) \cdot (s+5)}$



Asymptotes:

$$\angle\sigma_a = 180/3 + 360k/3 = 60, 180, 300, \quad \sigma_a = (-4-5)/3 = -3$$

$$tg^{-1}\frac{\omega}{\sigma+4} + tg^{-1}\frac{\omega}{\sigma+5} = \pi - tg^{-1}\frac{\omega}{\sigma}$$

$$\Rightarrow \frac{\dfrac{\omega}{\sigma+4} + \dfrac{\omega}{\sigma+4}}{1 - \dfrac{\omega}{\sigma+4}\cdot\dfrac{\omega}{\sigma+4}} = -\frac{\omega}{\sigma} \Rightarrow \frac{2\sigma+9}{\sigma^2+9\sigma+20-\omega^2} = -\frac{1}{\sigma}$$

$$3\sigma^2 + 18\sigma - \omega^2 + 20 = 0 \Rightarrow (\sigma+3)^2 - \frac{\omega^2}{3} = 9 - \frac{20}{3} \quad \textbf{(Hyperbola)}$$

$\omega = 0 \Rightarrow \sigma = \sqrt{7/3} - 3 = -1.47$ leaving point on the real axis

$\sigma = 0, \ \omega = \sqrt{20} = \pm 4.47$ intersect with imaginary axis

P10.8 Shown in Fig.P10.8 is the pole-zero map of a feedback control system in which the location of the real-axis pole is given by the variable ϕ. Determine the angle of emergence and entry into the complex poles and zeros when (a) ϕ=0 and (b) ϕ=-3 .Sketch the resulting root locus for each case.

Solution:

(a) $\phi=0$

emergence from pole

$$\psi = 180-135-90-45+72=-18$$

entry into zero

$$\psi = 180+135+135+108-90=108$$

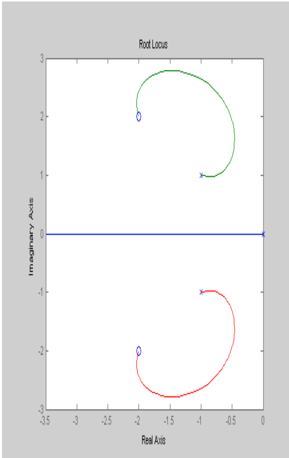
(b) ϕ=-3

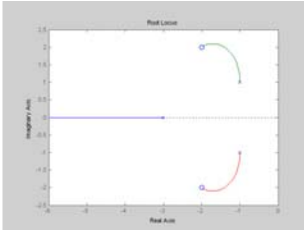
emergence from pole

$$\psi = 180-27-90-45+72=90$$

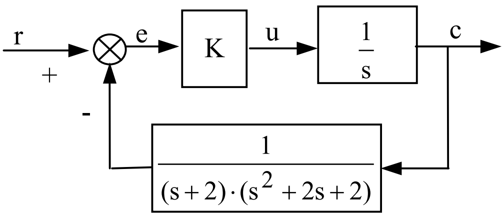
entry into zero

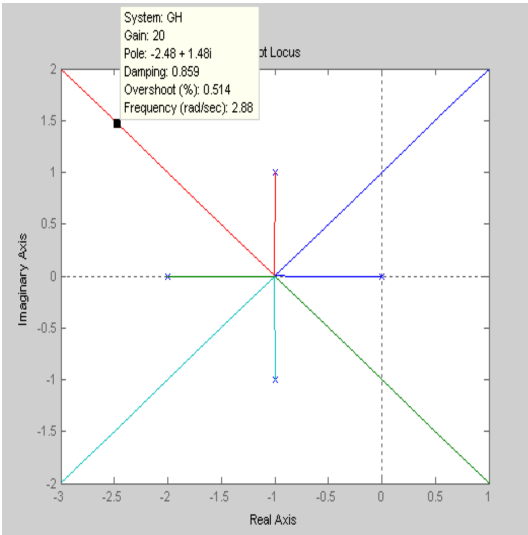
$$\psi = 180+63+135+108-90=36$$





P10.9 Plot the root locus for the feedback control system shown in Fig.P10.9, and determine the location of all roots for K=20.





Solution:

Conjugate poles locates on $-1\pm j\sqrt{2-1}$

Asymptotes:

$$\angle\sigma_a = 180/4+360k/4 = 45,135,225,315$$

$$\sigma_a = -4/4=-1$$

Leaving point on the real axis

S=-1 since $K=1>(1-\varepsilon)(1+\varepsilon)(1-\varepsilon^2)=1-\varepsilon^4$

Consider the location of root for gain K=20, x=y+1 and with

$$K=\left|(x-1)^2+(y-1)^2\right|\cdot\left|x^2+y^2\right|=\left|y^2+(y-1)^2\right|\cdot\left|(y+1)^2+y^2\right|.$$

$$=\left|2y^2-2y+1\right|\cdot\left|2y^2+2y+1\right|=(2y+1)^2-4y^2=4y^4+1.$$

$$y=0.707\cdot\sqrt[4]{K-1} \quad =1.48$$

It results in 4 roots of charaterstic equation for closed-loop:

-2.48-j1.48, -2.48+j1.48, 0.48-j1.48, 0.48+j1.48,

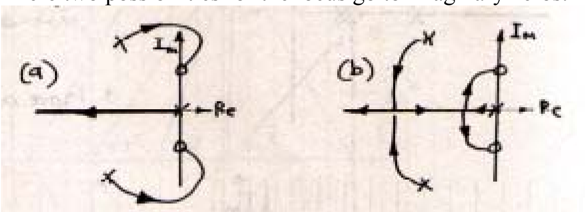
P11.5 A robot take television tubes from a convayer belt and places then in a test fixture. The vetical axis of the robot is controlled by a system with open-loop transfer function

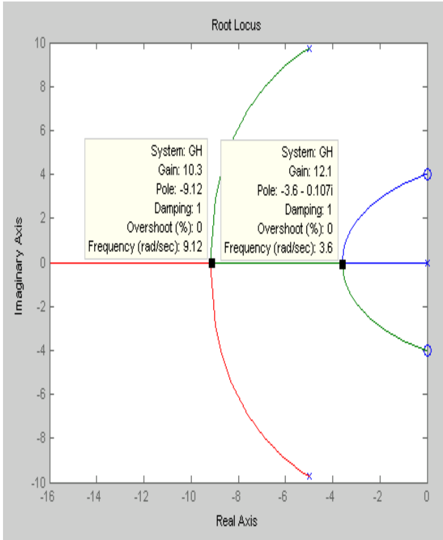
$$Gh(s) = \frac{K \cdot (s^2 + 16)}{s \cdot (s^2 + 10s + 120)}$$

Determine the "best" value of K to accomplish this task, stating reasons for the choice.

Solution:

There two possibilities for the locus go to imaginary zeros.





It depends the leaving points on the real axis. So from check gain on the real axis s=0 to -∞

$$K = -\frac{s \cdot (s^2 + 10s + 120)}{(s^2 + 16)}$$

$$\frac{dK}{ds} = \frac{(3s^2 + 20s + 120)(s^2 + 16) - 2s \cdot s \cdot (s^2 + 10s + 120)}{(s^2 + 16)^2}$$

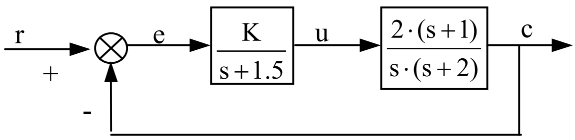
$s^4 - 78s^2 + 320s + 1920 = 0$ s=-3.5, s=-9.5

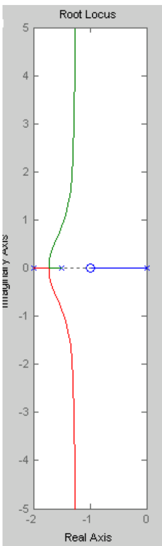
For robot conveyer, the best way is to avoid the overshoot.

So the damping ratio zeta=1, and take s=-3.5.The gain is about 12, $K = -\frac{-3.5 \cdot (12.25 - 35 + 120)}{(12.25 + 16)} = 12$

P11.8 Consider the control system shown in Fig.P11.8. Sketch the root locus and estimate the range of K for which the closed-loop poles satisfies the following design constraints:

a. Not more than 15% overshoot.
b. B. Dominant time constraint less than 2s.
c. C. Damped natural frequency of oscillation less than 2 rad/s.





Solution:

Asymptotes:

$\angle\sigma_a = 180/(3-1) + 360k/(3-1) = 90, 270$. $\sigma_a = (-1.5 - 2 + 1)/2 = -1.25$

$$K = -\frac{s \cdot (s+1.5) \cdot (s+2)}{2 \cdot (s+1)} \qquad \frac{dK}{ds} = \frac{(3s^2 + 7s + 2.5)(s+1) - s \cdot (s^2 + 3.5s + 2.5)}{(s+1)^2}$$

$2s^3 + 6.5s^2 + 7s + 2.5 = 0$ Leaving point on the real axis is s=-1.25

Consider the dominant effect of single pole on the real axis, is no any overshoot and damped natural frequency. Only condition to choice is time constant less than 2s.

So $\tau = \frac{1}{\varsigma\omega_n} < 2$, then the dominant pole should be $p = \varsigma\omega_n > 0.5$.

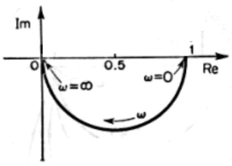
$$\Rightarrow \quad K > \frac{0.5 \cdot |0.5 - 1.5| \cdot |0.5 - 2|}{|0.5 - 1)|} = 1.5$$

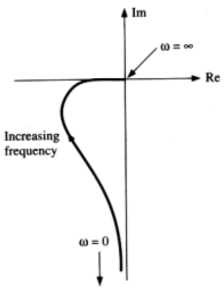
P12.1 For the following open-loop transfer functions, sketch the nyquist diagrams:

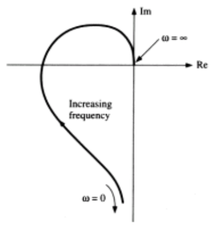
a. $GH(s) = \frac{K}{1+2s}$ b. $GH(s) = \frac{10}{s \cdot (1+4s)}$ c. $GH(s) = \frac{5}{s \cdot (s+2)^2}$ d. $GH(s) = \frac{4}{s^2 + s + 1}$

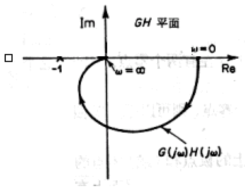
e. $GH(s) = \frac{5}{s \cdot (s+1)(s+2)}$

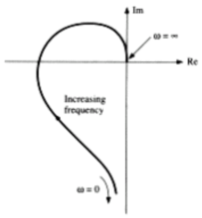
Solution:











P12.9 Figure P12.9 shows some experimentally determined open-loop frequency response plots. Match each plot with one of the transfer functions listed below:

a. $Gh(s) = \frac{1}{(1+sT_1)\cdot(1+sT_2)\cdot(1+sT_3)}$

b. $Gh(s) = \frac{K}{s \cdot (s+a)(s+b)}$

c. $Gh(s) = \frac{5}{s^3 \cdot (s+1)}$

d. $Gh(s) = \frac{1}{s}$

e. $Gh(s) = \frac{10}{(s+a)\cdot(s+b)\cdot(s+c)\cdot(s+d)}$

f. $Gh(s) = 1 + sT$

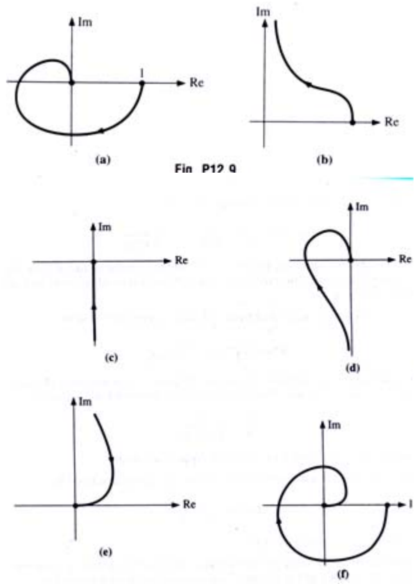


Fig P12.9

Solution:

a - (a), **b** – (d), **c** - (e), **d** –(c), **e** –(f), **f** - (b),

P13.1

a. In Fig. P13.3a, match each s-plane contour with the corresponding GH(s) contour.
b. In Fig. P13.3b, how many zeros are enclosed by the s plane contour?
c. In Fig. P13.3c, are any of the points A-E encircled by the contour? If so, how many and in which direction do the encirclements occur?

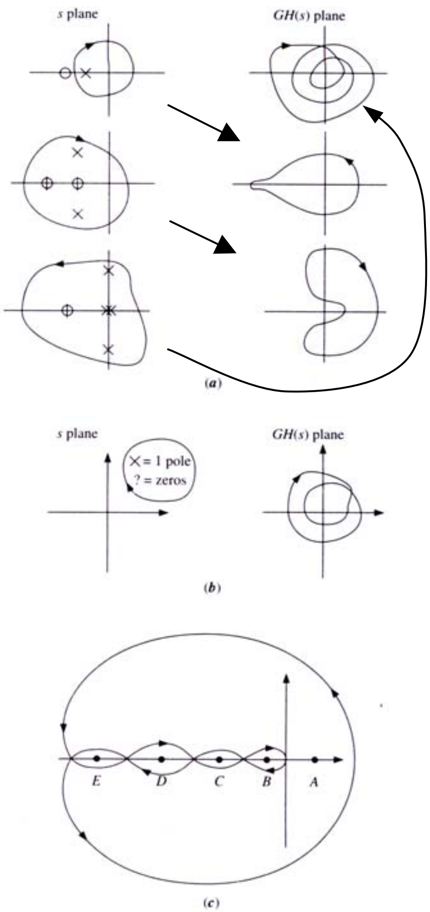


Fig. P13.3

Solution:
a. As arrowhead on the figure shown.
b. 3
c. A- 1 with clockwise
B- 0
C- 2 with anti-clockwise
D- 0 with clockwise
E- 2 with anti-clockwise

P13.4 For the following open-loop transfer functions, sketch the Nyquist diagrams for both positive and negative frequency and comment on the system stability:

a. $Gh(s) = \dfrac{K}{1+2s}$

b. $Gh(s) = \dfrac{10}{s\cdot(1+4s)}$

c. $Gh(s) = \dfrac{10}{s\cdot(s+2)^2}$

d. $Gh(s) = \dfrac{4}{s^2+s+1}$

e. $Gh(s) = \dfrac{5}{s\cdot(s+1)\cdot(s+2)}$

Solution:

a stable

b stable

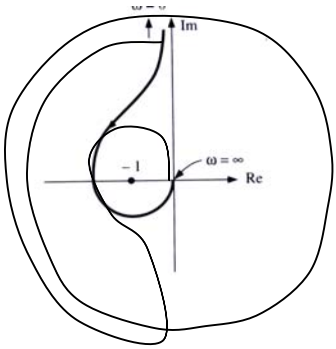
c Unstable

d stable

e stable: $\omega = \sqrt{2}\quad |GH(j\omega)| = \dfrac{5}{\sqrt{2}\cdot\sqrt{2+1}\cdot\sqrt{2+4}} = \dfrac{5}{6} < 1$

P13.7

Fig. P13.7 shows the open-loop frequency response of a system. Estimate the transfer function, and determine the stability of the closed-loop system.



Solution:

$$Gh(s) = \frac{K \cdot (s+a) \cdot (s+b)}{s^3}$$

It is stable, since Z=P+N = 2+ (-2) =0

P13.8 The open-loop frequency response shown in Fig.P13.8 was obtained experimentally. If the critical point (-1,0j) is located first at A, then at B, and finally at C, comment on the system stability. Reconsider stability of the modified contour shown in Fig. P13.6.

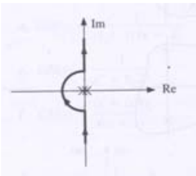


Fig. P13.6

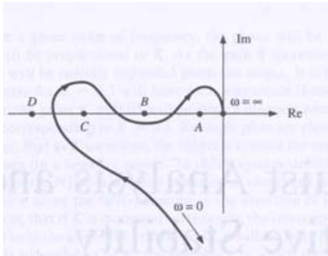


Fig. P13.8

Solution:

A-unstable, B- stable, C-unstable

P14.1 From the Nyquist diagram, find the maximum value of K for which the following systems are stable:

a. $Gh(s) = \dfrac{K}{s\cdot(s+a)}$ b. $Gh(s) = \dfrac{5K}{s^2\cdot(s+2)}$ c. $Gh(s) = \dfrac{K}{(s+1)^3}$ d. $Gh(s) = \dfrac{K}{(s+1)\cdot(s+2)^2}$

Solution:

a. Stable for any value of K

b. Unstable for any value of K.

c. Stable for K=8: $\angle GH(j\omega) = -3\cdot tg^{-1}\omega = 180 \Rightarrow \omega = tg60° = \sqrt{3} \Rightarrow K = K = (\sqrt{3+1})^3 = 8$

d. Stable for K=36

$$\angle GH(j\omega) = -tg^{-1}\omega - 2\cdot tg^{-1}\omega/2 \Rightarrow -\omega = \frac{\frac{\omega}{2}+\frac{\omega}{2}}{1-\frac{\omega^2}{4}} \Rightarrow \omega = \sqrt{8}$$

$$\Rightarrow |GH(j\omega)| = \frac{K}{\sqrt{8+1}\cdot(8+4)} \Rightarrow K=36$$

P14.2 A unity-feedback control system has the forward-path transfer function

$$G(s) = \frac{K}{s\cdot(s^2+s+9)}$$

Find the maximum value of K consistent with stability and check the result using Routh's method and by drawing the root locus.

Solution:

Obviously as ω=3, $\angle GH(j\omega)$=-180, $|GH(j\omega)| = K\cdot\dfrac{1}{3\cdot 3} = \dfrac{K}{9}$ => K<9

Gheck with Routh's: $G(s) = s^3 + s^2 + 9s + K$

so $K < 9$

P14.5 Fo the systems described by the Nyquistg diagrams shown in Fi. P14.5 estimate the gain margin and phase magin.

Solution:

a. PM>0 , GM= infinity

b. PM>0, GM>0

c. PM<0 , GM<0

d. PM>0 , infinity

e. PM>0 , GM= infinity

f. PM<0, >0 , GM<0, >0

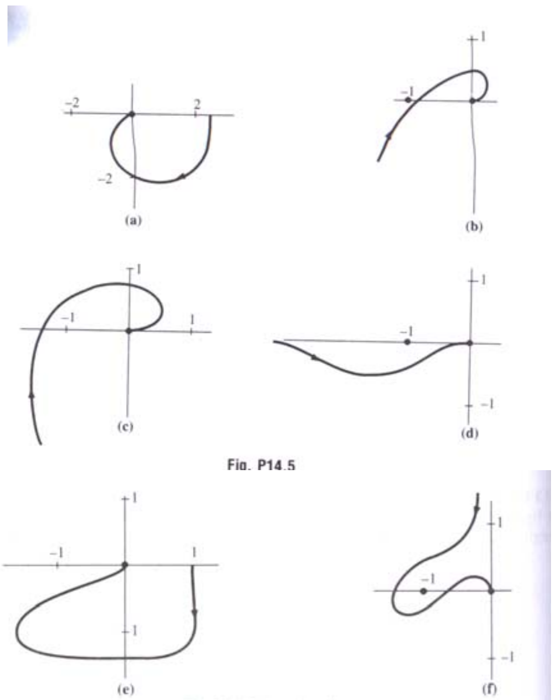


Fig. P14.5

Fig. P14.5 (continued)

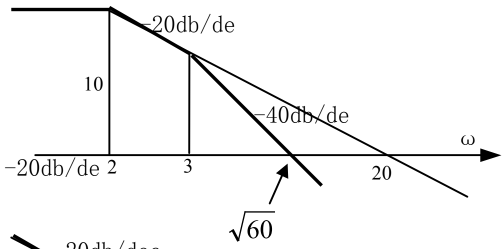
P15.1 Using straight-line approximations, plot the Bode diagram of the systems described by the following open-loop transfer function:

a. $Gh(s) = \dfrac{10}{(1+\frac{1}{3}s)(1+\frac{1}{2}s)}$ b. $Gh(s) = \dfrac{25(s+1)}{s\cdot(s+5)}$ c. $Gh(s) = \dfrac{16s}{(s+1)\cdot(s+2)}$

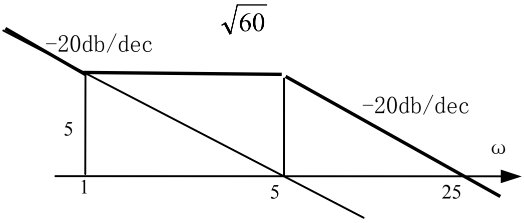
d. $Gh(s) = \dfrac{2\cdot(1+\frac{1}{10}s)\cdot}{s\cdot(1+\frac{1}{100}s)\cdot(1+\frac{1}{1000}s)\cdot(1+\frac{1}{2000}s)}$ .e. . $GH(s) = \dfrac{0.4\cdot(s+2)\cdot}{(s+0.2)\cdot(s+1)\cdot(1+\frac{1}{10}s)}$

Solution:

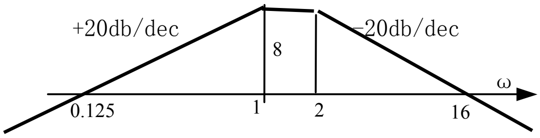
a. $GH(s) = \dfrac{10}{(1+s/3)(1+s/2)}$



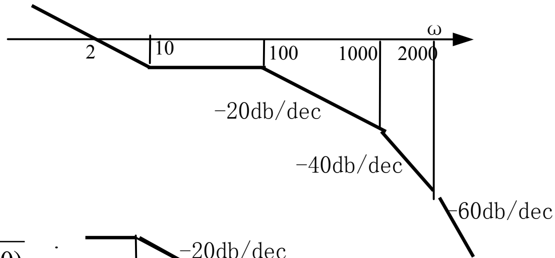
b $GH(S) = \dfrac{5\cdot(s+1)}{s\cdot(1+s/5)}$



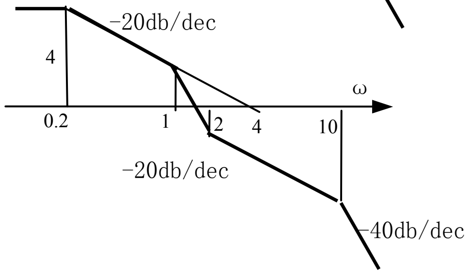
c.. $GH(s) = \dfrac{8s}{(s+1)\cdot(1+s/2)}$



d. $GH(s) = \dfrac{2\cdot(1+s/10)\cdot}{s\cdot(1+s/100)\cdot(1+s/1000)\cdot(1+s/2000)}$.



.e. . $GH(s) = \dfrac{4\cdot(1+s/2)\cdot}{(1+s/0.2)\cdot(1+s)\cdot(1+s/10)}$.



P15.2 The following systems contain second-order components. By estimating the magnitude and phase contributions from those elements, plot the Bode diagram for each system:

a. $GH(s)=\dfrac{15}{s^2+s+64}$ b. $GH(s)=\dfrac{1+0.2s}{s\cdot(s^2+0.5s+0.1)}$, c. $GH(s)=\dfrac{100\cdot(s+3)}{s^2\cdot(1+0.2s+0.01s^2)}$,

d. $GH(s)=\dfrac{20}{(s^2+2s+17)\cdot(s^2+3s+50)}$, e. $GH(s)=\dfrac{0.625\cdot(s^2+0.8s+4)}{s^2+4s+25}$,

Solution:

a. $$GH(s)=\frac{15/64}{\frac{1}{64}s^2+\frac{1}{64}s+1}$$

$$=\left|\frac{15}{\sqrt{(64-\omega^2)^2+\omega^2}}\right|\angle-\mathrm{arctg}\frac{\omega}{64-\omega^2}$$

b. $$GH(s)=\frac{10\cdot(1+0.2s)}{s\cdot(10s^2+5s+10)}$$

$$=\left|\frac{\sqrt{1+0.04\omega^2}}{\omega\cdot\sqrt{(1-\omega^2)^2+0.25\omega^2}}\right|\angle-90-\mathrm{tg}^{-1}\frac{0.5\cdot\omega}{1-\omega^2}+\mathrm{tg}^{-1}\frac{\omega}{5}$$

c. $$GH(s)=\frac{300\cdot(0.333s+1)}{s^2\cdot(1+0.2s+0.01s^2)}$$

$$=\left|\frac{100\sqrt{9+\omega^2}}{\omega^2\cdot(1+0.01\omega^2)}\right|\angle-180-\mathrm{tg}^{-1}\frac{0.2\cdot\omega}{1-0.01\omega^2}+\mathrm{tg}^{-1}\frac{\omega}{3}$$

d. $$GH(s)=\frac{2/85}{(\frac{1}{17}s^2+\frac{2}{17}s+1)\cdot(\frac{1}{50}s^2+\frac{3}{50}s+1)},$$

$$=\left|\frac{0.2}{\omega\cdot\sqrt{(17-\omega^2)^2+4\omega^2}\cdot\sqrt{(50-\omega^2)^2+9\omega^2}}\right|$$

$$\angle-\mathrm{tg}^{-1}\frac{2\omega}{17-\omega^2}-\mathrm{tg}^{-1}\frac{3\omega}{50-\omega^2}$$

e. $$GH(s)=\frac{0.1\cdot(0.25s^2+0.2s+1)}{0.04s^2+016s+1}$$

$$=\left|\frac{0.1\cdot\sqrt{(1-0.5\omega^2)^2+(0.2\omega)^2}}{\sqrt{(1-0.04\omega^2)^2+(0.16\omega)^2}}\right|$$

$$\angle-\mathrm{tg}^{-1}\frac{0.16\omega}{1-0.04\omega^2}+\mathrm{tg}^{-1}\frac{0.2\omega}{1-0.5\omega^2}$$

P16.1 For following systems, sketch the Bode diagram, and from the straight-line approximations to the gain and phase plots, estimate the maximum value of K for which the system is stable:

a. $Gh(s)=\dfrac{K}{s\cdot(s+1)\cdot(s+4)}$ b. $GH(s)=\dfrac{K}{s^2\cdot(1+s)}$ c. $Gh(s)=\dfrac{K\cdot s}{(s+2)^4}$

d. $Gh(s)=\dfrac{K}{s\cdot(s^2+2s+16)}$ .e. . $GH(s)=\dfrac{5K\cdot(1+s)\cdot}{s^2\cdot(1+s/3)^2}$

Solution:

a. $Gh(s)=\dfrac{K/4}{s\cdot(s+1)\cdot(s/4+1)}$

Let $PM=180-90-tg^{-1}\omega_c-tg^{-1}\omega_c/4=0$

=> $tg90°=\dfrac{\omega_c+\omega_c/4}{1-\omega_c\cdot\omega_c/4}=\infty$ => $\omega_c=2$

From $|GH(j\omega_c)|=1$, $\dfrac{K/4}{2\cdot\sqrt{2^2+1}\cdot\sqrt{0.5^2+1}}=1$ => K=20

b. $GH(s)=\dfrac{K}{s^2\cdot(1+s)}$, Always unstable.

c. $GH(s)=\dfrac{K\cdot s}{(s/2+1)^4}$

In order to let PM=0,

When $PM=180+90-4\cdot tg^{-1}\dfrac{\omega_c}{2}=0$,

$$\omega_c=2\cdot tg\frac{270°}{4}=4.85$$

From $|GH(j\omega_c)|=1$ => $K=\dfrac{(2.426^2+1)^2}{4.85}=9.7$

d. $GH(s)=\dfrac{K/16}{s\cdot(s^2/16+s/8+1)}$.

PM=0 => $\omega_c=4$

From $|GH(j\omega_c)|=1=\dfrac{K/16}{4\cdot(1/2)}$ => K=32

e. . $GH(s)=\dfrac{5K\cdot(1+s)\cdot}{s^2\cdot(1+s/3)^2}$

Let $PM=180-180+tg^{-1}\omega_c-2\cdot tg^{-1}\omega_c/3=0$

By strait line approximations to the phase plot, suppose

=> $\omega_c=\dfrac{\omega_c/3+\omega_c/3}{1-\omega_c\cdot\omega_c/9}$ => $\omega_c=\sqrt{3}$

From $|GH(j\omega_c)|=1$, $\dfrac{5K\cdot\sqrt{1+3}}{3\cdot(1+1/3)}=1$ => K=2/5=0.4

P16.2 Estimate for each system given below the gain margin and phase margin:

a. $GH(s)=\dfrac{15\cdot(s+2)}{s^2\cdot(s+9)}$, b. $GH(s)=\dfrac{20}{s^2\cdot(s+5)\cdot(s^2+2s+4)}$, c. $GH(s)=\dfrac{0.1\cdot(s^2+2s+9)}{s^2\cdot(1+s/10)^2}$,

d. $GH(s)=\dfrac{2000\cdot(s+1)}{s\cdot(s+2)\cdot(s+5)\cdot(s+10)\cdot(s+20)}$, e. $GH(s)=\dfrac{0.5\cdot(s+10)\cdot(s+20)}{s^2\cdot(s+1)\cdot(s+100)}$,

Solution:

a. $GH(s)=\dfrac{10/3\cdot(s/2+1)}{s^2\cdot(s/9+1)}$,

$\omega_c=\sqrt{10/3}=1.83$,

$PM=tg^{-1}\dfrac{1.83}{2}-tg^{-1}\dfrac{1.83}{9}=42.5\text{-}11.5\text{=}31$

b. $GH(s)=\dfrac{1}{s^2\cdot(s/5+1)\cdot(s^2/4+s/2+1)}$,

$\omega_c=1$, $PM=-tg^{-1}\dfrac{1}{5}-tg^{-1}\dfrac{0.5}{1-0.25}\text{=-}45$

c. $GH(s)=\dfrac{0.9\cdot(s^2/9+2s/9+1)}{s^2\cdot(1+s/10)^2}$,

$\omega_c=\sqrt{0.9}\text{=}0.95$,

$PM=tg^{-1}\dfrac{2\cdot 0.95/9}{1-0.95^2/9}-2\cdot tg^{-1}\dfrac{0.95}{10}\text{=}2.3$

d. $GH(s)=\dfrac{s+1}{s\cdot(s/2+1)\cdot(s/5+1)\cdot(s/10+1)\cdot(s/20+1)}$,

$\omega_c=1.5$,

$PM=tg^{-1}1-tg^{-1}\dfrac{1}{2}-tg^{-1}\dfrac{1}{5}-tg^{-1}\dfrac{1}{10}-tg^{-1}\dfrac{1}{20}$

$=45-26.6-11.3-5.7-2.9=-1.6$

e. $GH(s)=\dfrac{(s/10+1)\cdot(s/20+1)}{s^2\cdot(s+1)\cdot(s/100+1)}$,

$\omega_c=1$,

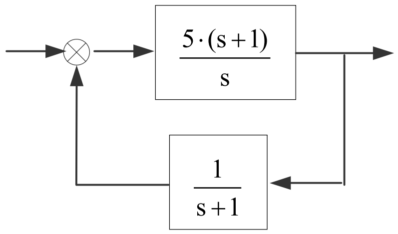
$PM=-tg^{-1}1+tg^{-1}0.1+tg^{-1}0.05-tg^{-1}0.01$

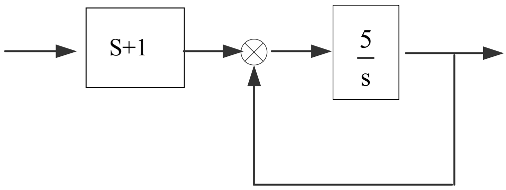
$=-45-5.7-2.86-0.57=-54.13$

P17.7 Consider the system described by the open-loop transfer function element GH(s)=5/s and H(s)=1/(s+1) Draw the Bode diagram of the open-loop transfer function and write down an expression for the unit impulse response of the closed-loop system.

Solution

GH(s)=5/s, and H(s)=1/(s+1), then $G(s)=\dfrac{5\cdot(s+1)}{s}$

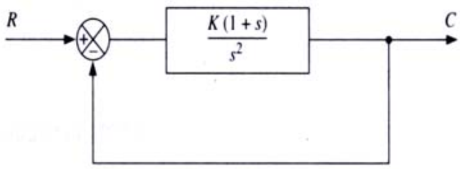




By block diagram algebra, $\dfrac{Y}{R}=\dfrac{5}{s+5}\cdot(s+1)=\dfrac{5s+5}{s+5}=\dfrac{5s+25-20}{s+5}=5-\dfrac{4\cdot 5}{s+5}$.

For unit impulse input, $y(t)=5\cdot\delta(t)-4e^{-5t}$

P17.8 Sketch the root locus of the system shown in Fig.P17.8 and determine the value of K needed to give the dominant closed-loop poles a damping ratio of ζ =0.707. Estimate the closed–loop unit step response of the system by potting the Bode diagram of the open-loop transfer function for this value of K, and discuss the two results. Compare the result with direct analysis of the closed-loop transfer function.



Solution

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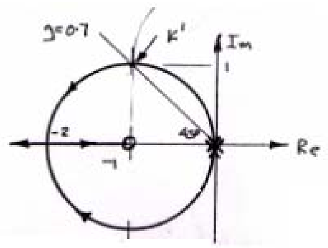
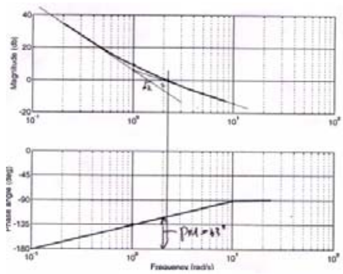




By root locus, the K should be calculated as follow:

$K=\sqrt{2}\cdot\sqrt{2}/1=2$

By Bode diagram, ζ =0.707 => PM=63

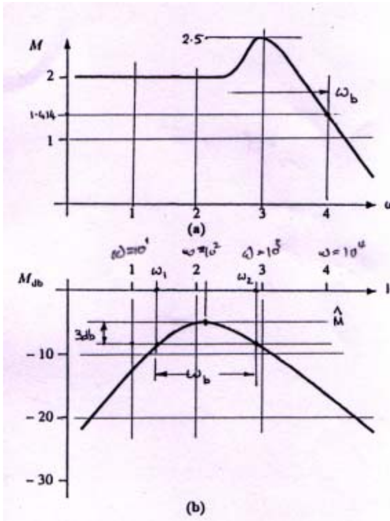
by $PM=180-90-tg^{-1}(\omega_c/1)$ => $\omega_c=2$

by K= $(\sqrt{K})^2=1\cdot\omega_c$ =2

$\dfrac{C}{R}=\dfrac{K(1+s)}{s^2+Ks+K}=\dfrac{K(1+s)}{s^2+2\cdot\sqrt{K}/2\cdot\sqrt{K}\cdot s+K}$ $\omega_n=\sqrt{K}$, $\varsigma=\dfrac{\sqrt{K}}{2}$ If ζ =0.707, then the K=2.

P18.1 Figure P18.1 shows the magnitude part of the frequency response of different systems. For each case determine the bandwidth, the size of the resonance peak, and the frequency at which the peak occurs.

Solution



(a) $-3\text{db} = 0.707 \cdot \text{DV} = 1.414$

$\omega_b = 4\text{rad/s}$

$M_p = 2.5\omega_b \qquad \omega_r = 3\text{rad/s}$

(b) take –3db with Mp

$\omega_1 = 28.2$

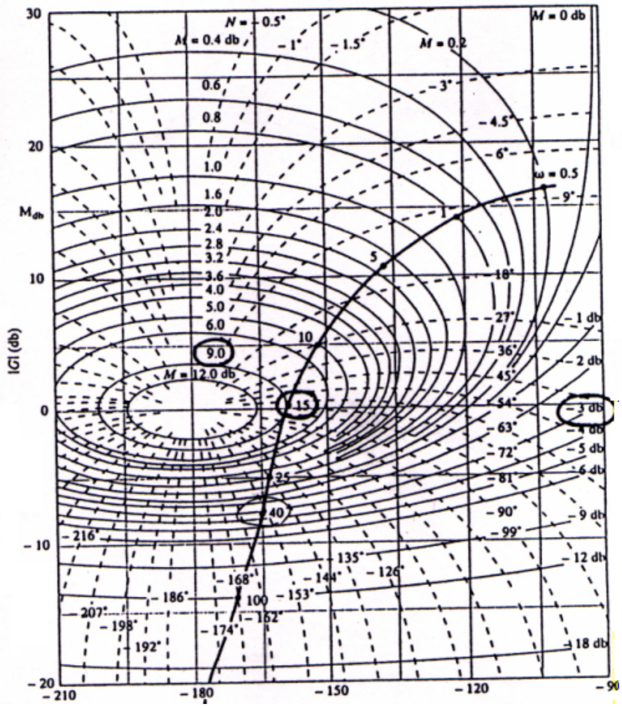
$\omega_1 = 794.3$

$\omega_b = \omega_2 - \omega_1 = 766$

$M_p = -5\text{db} \qquad \omega_r = 105\text{rad/s}$

P18.7 Consider the system shown in Fig. P18.7. Estimate the peak time and percentage overshoot for the closed-loop system if the input is a unit step.

Solution



From Mr=9db => $M_r = 2.8$

$\omega_r = 15\text{rad/s}$

By Fig18.9, or formulation

$$M_p = \frac{1}{2\varsigma \cdot \sqrt{1-\varsigma^2}}$$

=> $\zeta = 0.16$

From $\omega_r = 15\text{rad/s}$

By Fig18.10, or formulation

$$\omega_r = \omega_n \cdot \sqrt{1-2\varsigma^2}$$

=> ω_n=

take –3db with M, $\omega_b = 40$

Form Fig 18.10 or formulation

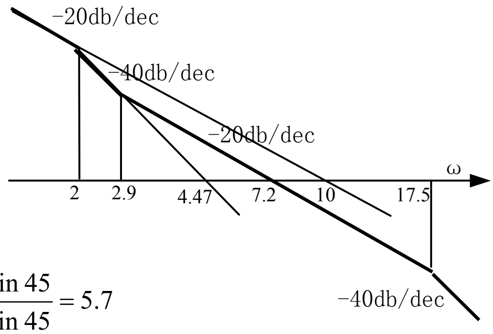
$$\omega_b = \omega_n \cdot \left(1-2\zeta^2 + \sqrt{2-4\zeta^2+4\zeta^4}\right)^{1/2}$$

$\omega_n = 26.7 \Rightarrow \omega_d = 26.3$ => $T_p = \pi / \omega_{d=0.12s}$, From Fig4.17 PO=60%

P19.2 Consider the unity-feedback system in Fig.P19.2. Design a phase lead compensation network that give the closed-loop system: GH(s)=K/(s*(1+0.5s))

a. A phase margin of 50^o .

b. A velocity error of less than 10%.



Solution

By requirement e_{ss}=10% for ramp input, => $K=K_v=10$

New $\omega_c = \sqrt{K \cdot 2} = \sqrt{20} = 2\sqrt{5} = 4.47$

$$PM0 = 180 - GH(\omega_c) = 180 - 90 - tg^{-1}\frac{4.47}{2} = 24.2$$

By requirement PO=10% => ζ =0.6 => PM =60 =>

$$\phi_m = PM - PM0 = 60 - 24 = 36 \text{ Take } 45, \quad \alpha = \frac{1+\sin 45}{1-\sin 45} = 5.7$$

New $\omega_m = \omega_c \cdot \sqrt{\sqrt{6}} = 4.47 \cdot 1.6 = 7.15$

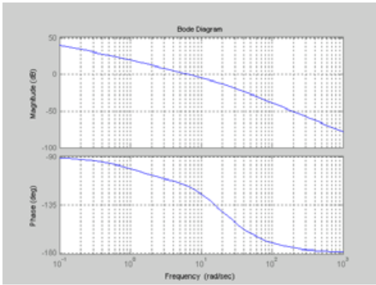
$\omega_p = \omega_m \cdot \sqrt{6} = 7.15 \cdot 2.45 = 17.5$,

$\omega_z = \omega_m / \sqrt{6} = 7.15 / 2.45 = 2.92$,

$$\alpha T = \frac{1}{\omega_z} = 0.34, \quad T = 0.057$$

$$G_c(s) = \frac{1+\alpha Ts}{1+Ts} = \frac{1+0.34s}{1+0.057s} ,$$

$$GH(s)G_c(s) = \frac{10 \cdot (1+0.34s)}{s \cdot (1+0.5s) \cdot (1+0.057s)}$$

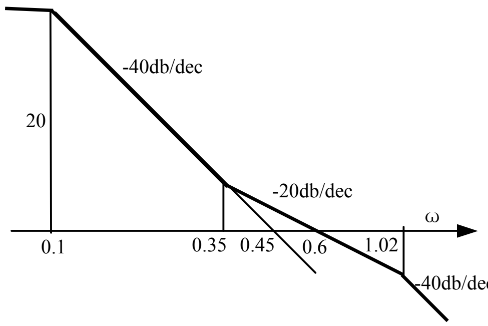


P19.3 Design a phase lead cascade compensation filter that provides the unity-feedback system

$$GH(s) = \frac{20}{(1+10s)^2}$$

with a percentage overshoot to a unit step of less than 15%.

Solution



$\omega_c = \sqrt{20} \cdot 0.1 = 0.45$

$$PM0 = 180 - GH(\omega_c) = 180 - 2 \cdot tg^{-1} 4.5 = 35°$$

PO=15% => ζ =0.6 => PM =60°

$$\phi_m = PM - PM0 = 60 - 35 = 25 \text{ Take } 30$$

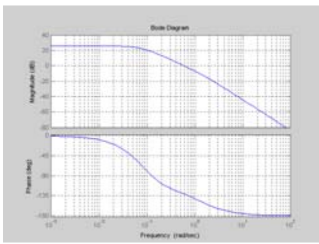
$$\alpha = \frac{1+\sin 30}{1-\sin 30} = 3$$

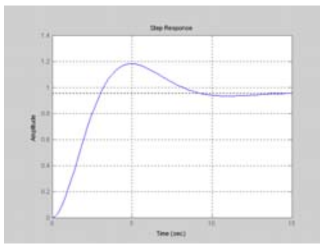
New $\omega_m = \omega_c \cdot \sqrt{\sqrt{3}} = 0.45 \cdot 1.316 = 0.6$

$\omega_p = \omega_m \cdot \sqrt{3} = 0.6 \cdot 1.732 = 1.02$, $\omega_z = \omega_m / \sqrt{3} = 0.6 / 1.732 = 0.35$

$$G_c(s) = \frac{1+\alpha Ts}{1+Ts} = \frac{1+2.9s}{1+0.97s}$$

$$GH(s)G_c(s) = \frac{20 \cdot (1+s/0.35)}{(1+10s)^2 \cdot (1+s/1.02)}$$





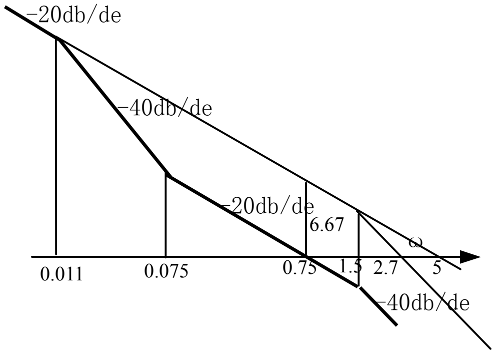
P20.2 A unity feedback control system has the open-loop transfer function

$$GH(s) = \frac{K}{s \cdot (1+0.67s)}$$

Design a suitable phase lag compensation filter that ensures that the closed-loop system meets the following performance requirements:

a. A velocity error no more than 20%..

b. A maximum percentage overshoot to a step of 10%



Solution

By requirement e_{ss}=20% for velocity error => $K=K_v=5$

By requirement PO=10% => ζ =0.6 => PM =60°

Draw block diagram,

the integrator forms a line with slope of -20db/dec.

$\omega_0 = 5 \;\; => \;\; \omega_c = \sqrt{5 \cdot 1/0.67} = \sqrt{7.5}$ =2.74

$$PM = 180 - GH(\omega_c) = 180 - 90 - tg^{-1}\frac{2.45}{1.5} < 60°$$

Select new ω'$_c$, let PM=60° => $\omega_c = 1/(2 \cdot 0.67) = 0.75$

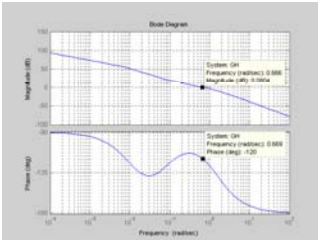
=> $\beta = GH(\omega'_c) = GH(0.75) = 5/0.75 = 6.67$

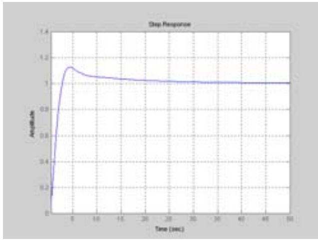
$\omega_z = \omega'_c/10 = 0.75/10 = 0.075$

$\omega_p = \omega_z/\beta = 0.075/6.67 = 0.011$,

$$G_c(s) = \frac{1+Ts}{1+\beta Ts} = \frac{1+s/0.075}{1+s/0.011}$$

$$GH'(s) = \frac{10 \cdot (1+s/0.075)}{s \cdot (1+0.67s) \cdot (1+s/0.011)}$$





P20.4 Design a phase lag cascade compensation filter Gc(s) for open-loop system

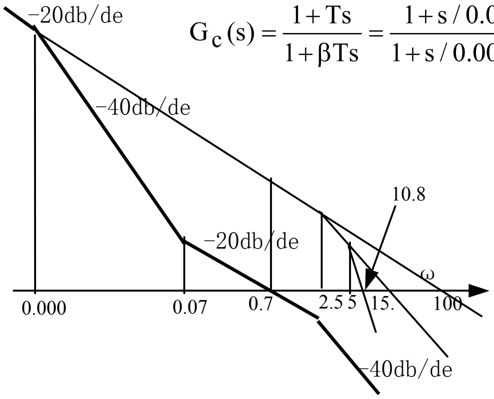
$$GH(s) = \frac{K}{s \cdot (1+0.2s)(1+0.4s)}$$

such that the closed-loop system has a damping ratio ζ =0.7 and a velocity error Kv=100. What is the estimate settling time of the compensated-system output when the input is a unit step? Sketch the root locus of the uncompensated and compensated systems.

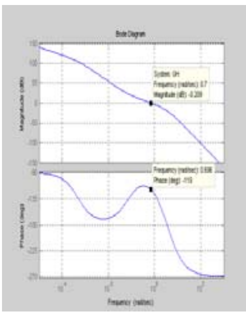
Solution

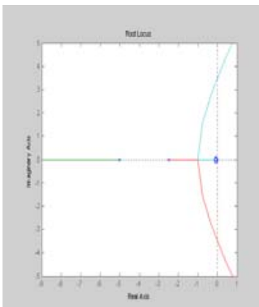
By requirement $K=K_v$=100, ζ =0.7 => PM =65. => new $\omega_c = 0.7$ for PM

$\beta = GH(\omega'_c) = GH(1) = (100/0.7) = 143 \;\; \omega_z = 0.7/10 = 0.07$, $\omega_p = \omega_z/\beta = 0.0005$



$$G_c(s) = \frac{1+Ts}{1+\beta Ts} = \frac{1+s/0.07}{1+s/0.0005} \qquad GH'_c(s) = \frac{100}{s \cdot (1+0.2s)(1+0.4s)} \cdot \frac{1+s/0.07}{1+s/0.0005}$$

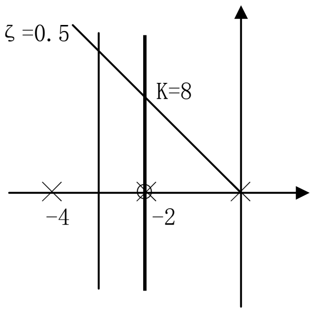




P21.2 A unity feedback comtrol system has the open-loop transfer function

$$GH(s) = \frac{40}{(s+4)\cdot(s+2)}$$

Use the root locus method to design a suitable PI controller such that the system has zero steady-state position error and the dominant closed-loop poles have a damping ratio of 0.5. Estimate the percentage overshoot, peak time, and 2% settling time of the closed-loop, unit step response.



Solution

By requirement e_{ss}=0 for position error => Type I

By requirement ζ =0.5 => PM >45

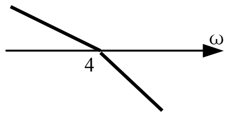
Bode form $GH(s) = \dfrac{5}{(1+s/4)\cdot(1+s/2)}$

Using a PI controller, the transfer function become

$$GH'(s) = \frac{5\cdot K_1\cdot(T_1 s+1)}{s\cdot(1+s/4)\cdot(1+s/2)}$$

Since there is a integrator in the GH'(s), then ess for position error will be zero.

In order to meet the need of ζ =0.5, set T_1=1/2=0.5, and $5K_1$=4 , that is K_1=0.8

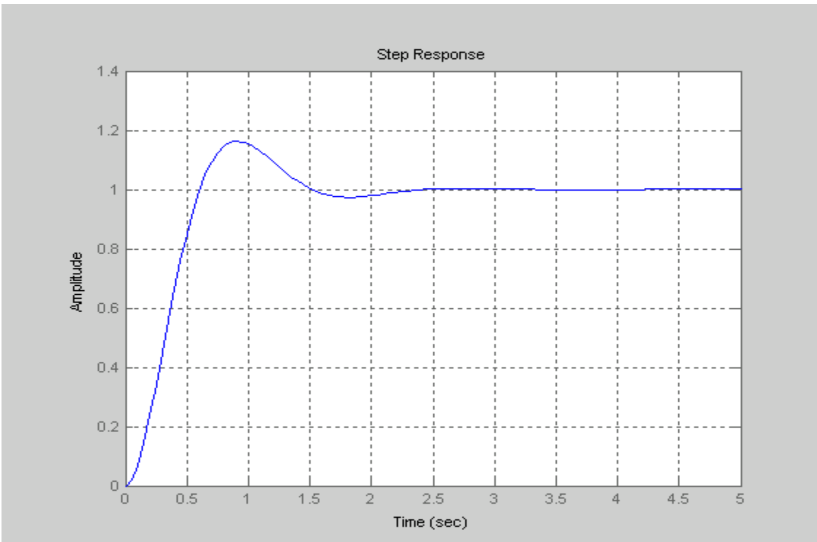


$$GH'(s) = \frac{4\cdot(0.5s+1)}{s\cdot(1+s/4)\cdot(1+s/2)}, \quad Gc(s) = \frac{0.8\cdot(0.5s+1)}{s}$$

ζ =0.5 => PO=16%,

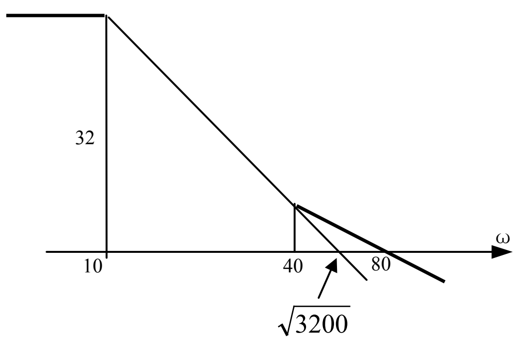
$$t_s = \frac{4}{\varsigma\omega_n} = \frac{4}{0.5\cdot\omega_c\ /(1.1-0.67\varsigma)} = \frac{8}{4/0.78} = 1.6s$$

peek time $t_p = \dfrac{\pi}{\omega_d} = \dfrac{3.14}{(1-0.5^2)\cdot 4/0.78} = 0.8s$



P21.6 For the unity-feedback system described by the open-loop transfer function

$$GH(s) = \frac{5}{(s+10)^2}$$



Design a PD controller such that the closed-loop system has a damping ratio of ζ=0.7 and the 2 settling time is less than 0.1s.

Solution

Using PD controller the new GH becomes

$$GH'(s) = \frac{0.05 \cdot K_1 \cdot (T_1 s + 1)}{(1 + s/10)^2}$$

By requirement, ζ=0.7, ts=0.1s => ω_c=8/ts=80

ζ=0.7=> PM=63° => $1/T_1 = \omega_c/2$ =40

$$K = 0.05 \cdot K_1 = \left(\frac{\sqrt{3200}}{10}\right)^2 = 32$$

=> T_1=1/40=0.025, K_1=32/0.05=640

$$GH'(s) = \frac{32 \cdot (1 + s/40)}{(1 + s/10)^2}$$ Gc(s)=640 (0.025s+1)